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Vision-based reduced-order adaptive control of aerial continuum manipulation systems

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ABSTRACT

In this paper, a vision-based control method for aerial continuum manipulation systems (ACMSs) is proposed. The objective of this paper is to design a proper controller for highly nonlinear and coupled aerial continuum manipulation system in the presence of uncertainties. This paper contributes by outlining the design of the first vision based adaptive control for ACMSs which circumvents the need for a priori knowledge of system dynamic model and allows robot and target positioning uncertainties. Using vision feedback in combination with adaptive control method enables effective treatment of nonlinearities, coupling and uncertainties present in typical ACMSs. Designing controller for such nonlinear systems in the absence of exact system model is quite challenging. To this end, reduced order adaptive method is proposed which is a computationally efficient technique. The simulation results demonstrate the performance of the proposed control method.

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1. Introduction

Owing to their extensive applications, many recent works have focused on developing effective aerial manipulation systems (AMSs) [1]. As a result, a variety of base platforms [2–4] and arms with different number of degrees-of-freedom [1,5] have been proposed to develop efficient and practical aerial manipulation systems. An emerging trend in aerial manipulation also includes reforming the arm types from rigid-link robots to soft [6] and continuum ones [7] to improve their operational capabilities. Despite significant previous work on modeling and control of AMSs with rigid-links, the related research work on AMSs with soft and continuum robots is in its infancy.

Fig. 1 shows a schematic of AMS with a continuum robot. Such integrated systems are called aerial continuum manipulation systems (ACMSs) [8]. The advantages of ACMSs originate from their biological counterparts such as trunks and tongues. Examples include higher interaction safety, dexterity, flexibility, built-in compliance (to enable operation in unstructured environments and within obstacles) and ease of their adaptation to objects' shapes in their grasp and manipulation tasks [8]. The aforementioned advantages could be put into practice if the pending complicated issues related to modeling, control and planning of ACMSs are addressed [7,8]. The focus of this paper is placed on designing a high-performance controller for ACMSs. While conventional aerial manipulation systems are still investigated, the replacement of conventional rigid-link arms with their continuum counterparts adds to the complexity of control design.

Dynamic control of continuum arm is challenging issue by itself due to the significant level of nonlinearities embodied in their dynamics. Majority of the existing controllers have been designed for quasi-static and kinematic solutions. Limited model-based works exist in the literature on dynamic control of sole continuum arm [9–11]. These methods are not computationally efficient which makes them inappropriate for real-time implementation and practical usage. To overcome these problems in dynamic control of continuum arm, modeless control methods have attracted attentions. For instance, learning-based control approaches have been adopted without any need of knowledge in dynamics of the system [12,13]. However, learning-based controllers need significant training effort for each aerial mission which can result in additional difficulties for performing aerial manipulation missions. The problem is further aggravated by attaching the continuum arm to unmanned aerial vehicles (UAVs) [7]. For example, such an integration will lead to the coupling nonlinearities and increased dynamic uncertainties, adversely affecting the control performance [14]. The structural nature of continuum arms also makes them more susceptible to flight-induced disturbances such as wind gusts and downwash effects. Due to the complicated dynamic models

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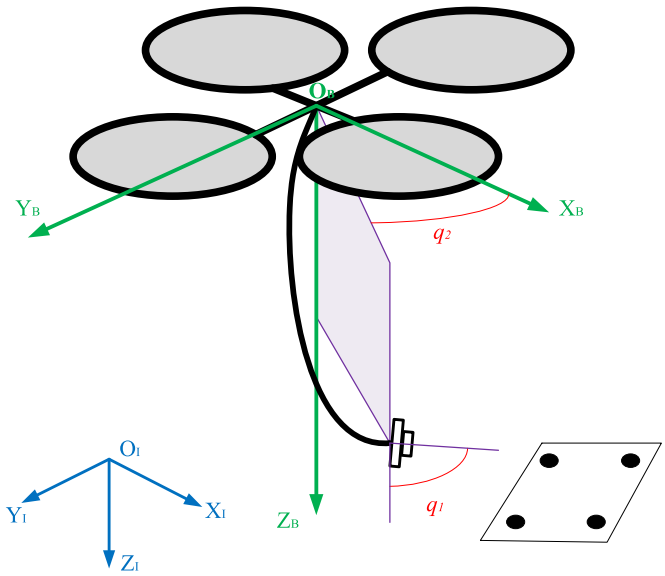

Fig. 1. Aerial continuum manipulation system.

of ACMSs, model-based control approaches need to overcome the computational load issue before becoming practical. Therefore, design of dynamic controllers with no or reduced need to dynamic models of ACMSs is a high priority research topic. The previous work [7] was the first published work on modeling and control of ACMSs. However, the previous approach was model-based and required a good knowledge of the system model. Additionally, the previous work relied on decoupled system dynamics, omitting the coupled dynamic terms and simplifying the interaction forces between the UAV and arm. The previous control method also had limited degree-of-freedom (DOFs) and was computationally expensive. Furthermore, the previous attempt lacked any visual servo control scheme, preventing the ACM to operate in high levels of position uncertainties, such as positioning relative to a moving object.

In this paper, a cascaded control approach is taken to address the aforementioned issues. First, a reduced-order adaptive control approach is developed and proposed for the inner loop to circumvent the need for system dynamics. In the proposed method, the system dynamics is estimated on-the-fly, i.e., during the control process [15]. The second controller in the outer-loop is a visual-servo control system. The integration of visual servoing (VS) is motivated by the existence of significant levels of uncertainties and disturbances in any ACMS operation that might negatively impact positioning accuracy of the end-effector for any manipulation task. Examples of common position uncertainties include joint slippages, gear backlashes, links deflection (which is significantly pronounced in continuum robots), model imperfections, and target pose inaccuracies due to its motion or improper placement [16,17]. Another source of positioning challenge would originate from the base of the arm, i.e., UAVs, which, due to their structural and motion flexibility, pose added positioning uncertainties, when compared to ground-based arms [18,19]. Accordingly, the integration of VS seems a natural choice for enhancing the positioning accuracy of the robot [20].

Visual servoing systems can be categorized based on camera configuration and/or control architecture. Depending on the position of the camera, there are two main configurations namely eye-in-hand (EIH) and eye-to-hand (ETH) [17,21]. In the EIH configuration, a dynamic camera is fixed at the end-effector and moves with the end-effector to observe a fixed or moving object; while in the ETH configuration, a static camera is fixed and observes the end-effector and reference object. From the control perspective, two main visual servoing methods are image-based visual servoing (IBVS), and position-based visual servoing (PBVS) [16,17,22]. The IBVS scheme has some advantages over PBVS method. One advantage is that the geometry of target is not needed in the IBVS method. Moreover, since IBVS approach just uses image feature information, it is more robust to calibration errors and image noise compared to the PBVS method. Due to the above-mentioned advantages, an eye-in-hand IBVS scheme is adopted here for ACM control. IBVS scheme has been utilized in aerial missions such as flexible aerial refueling [23] and impedance control of a dual arm aerial manipulator [24]. Furthermore, an enhanced IBVS method is proposed in [25] to guarantee visibility in quadrotor visual servoing control. However, despite the VS advantages, no visual servo control scheme has been designed for ACMSs yet.

This paper contributes by designing the first vision based adaptive control for ACMSs which circumvents the need for a priori dynamic model of system and allows robot and target positioning uncertainties. The performance of the proposed control method is verified through the simulation results.

The structure of the paper is as follows. System modeling is discussed in section 2. Section 3 includes the description of the control architecture, followed by the development of reduced-order adaptive control scheme in section 4. Section 5 explains visual servo control design. Simulation results are given in section 6, followed by the conclusions provided in section 7.

2. Modeling

In this section, modeling of ACMS is provided to enable dynamic simulations of the proposed control approach. The approach is based on Euler-Lagrange formulation. Here, it is assumed that a tendon-driven continuum arm is attached to the center of a multi-rotor UAV (Fig. 1).

The rotation matrix, $\boldsymbol{R}_{\mathrm{B}}$, describes the rotation between the inertial reference frame $\mathcal{F}_{\mathrm{I}}$ with the origin $\mathrm{O_I}$ and the UAV's body frame $\mathcal{F}_{\mathrm{B}}$ with the origin O_{B} as

$$\boldsymbol{R}_{\mathrm{B}} = \begin{bmatrix} \mathrm{C}\psi & -\mathrm{S}\psi & 0 \\ \mathrm{S}\psi & \mathrm{C}\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathrm{C}\theta & 0 & \mathrm{S}\theta \\ 0 & 1 & 0 \\ -\mathrm{S}\theta & 0 & \mathrm{C}\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \mathrm{C}\varphi & -\mathrm{S}\varphi \\ 0 & \mathrm{S}\varphi & \mathrm{C}\varphi \end{bmatrix}, \tag{1}$$

where C and S denote cosine and sine function, respectively. The position of the UAV in the inertia frame is indicated by $\boldsymbol{p}_{\mathrm{B}}$ whereas, its attitude is shown by $\boldsymbol{\phi}_B = \begin{bmatrix} \varphi & \theta & \psi \end{bmatrix}^{\mathrm{T}}$ containing roll, pitch, and yaw angles, respectively. The linear velocity of UAV in inertial reference frame, $\dot{\boldsymbol{p}}_{\mathrm{B}}$, can be expressed as

$$\dot{\boldsymbol{p}}_{\mathrm{B}} = \boldsymbol{R}_{\mathrm{B}} \dot{\boldsymbol{p}}_{\mathrm{B}}^{\mathrm{B}}, \tag{2}$$

where $\dot{\boldsymbol{p}}_{\mathrm{B}}^{\mathrm{B}}$ denotes the velocity of UAV in its body frame. Similarly, the angular velocity of UAV in $\mathcal{F}_{\mathrm{I}}$, $\boldsymbol{\omega}_B$, can be determined as

$$\boldsymbol{\omega}_B = \boldsymbol{T}_B \dot{\boldsymbol{\phi}}_B, \tag{3}$$

where $\dot{\boldsymbol{\phi}}_B$ is derivative of Euler angles and $\boldsymbol{T}_{\mathrm{B}}$ is defined as

$$\boldsymbol{T}_{\mathrm{B}} = \begin{bmatrix} 1 & 0 & -\mathrm{S}\theta \\ 0 & \mathrm{C}\varphi & \mathrm{S}\varphi\mathrm{C}\theta \\ 0 & -\mathrm{S}\varphi & \mathrm{C}\varphi\mathrm{C}\theta \end{bmatrix}. \tag{4}$$

Therefore, the angular velocity of UAV in $\mathcal{F}_{\mathrm{B}}$ can be calculated as

$$\begin{aligned} \boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{B}} &= \boldsymbol{R}_{\mathrm{B}}^{\mathrm{T}} \boldsymbol{\omega}_{\mathrm{B}} = \boldsymbol{R}_{\mathrm{B}}^{\mathrm{T}} \boldsymbol{T}_{\mathrm{B}} \dot{\boldsymbol{\phi}}_{\mathrm{B}} = \boldsymbol{Q}\, \dot{\boldsymbol{\phi}}_{\mathrm{B}}, \\ \boldsymbol{Q} &= \boldsymbol{R}_{\mathrm{B}}^{\mathrm{T}} \boldsymbol{T}_{\mathrm{B}}. \end{aligned} \tag{5}$$

Assuming a constant-curvature model for the arm, the end-effector coordinates can be represented by two angles [26]. By considering $\boldsymbol{q} = \begin{bmatrix} q_1 & q_2 \end{bmatrix}^{\mathrm{T}}$ as angles of end effector for continuum arm, with q_1 denoting the vertical angle with respect to UAV and q_2 showing horizontal angle, the generalized states of the ACMS can be expressed as:

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{p}_{\mathrm{B}}^{\mathrm{T}} & \boldsymbol{\phi}_B^T & q_1 & q_2 \end{bmatrix}^{\mathrm{T}}. \tag{6}$$

The position of each point of the continuum arm in $\mathcal{F}_{\mathrm{I}}$ can be defined as

$$\boldsymbol{p}_{\mathrm{s}} = \boldsymbol{p}_{\mathrm{B}} + \boldsymbol{R}_{\mathrm{B}} \boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}, \tag{7}$$

where $\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}$ is the relative position of each section (point) of the arm in UAV body frame and can be determined as:

$$\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}} = \begin{bmatrix} \frac{l}{q_1}\left(1 - \mathrm{C}\left(\frac{sq_1}{l}\right)\right)\mathrm{C}q_2 \\ \frac{l}{q_1}\left(1 - \mathrm{C}\left(\frac{sq_1}{l}\right)\right)\mathrm{S}q_2 \\ \frac{l}{q_1}\mathrm{S}\left(\frac{sq_1}{l}\right) \end{bmatrix}, \tag{8}$$

where l and s denote the length of the continuum arm and a section of it, respectively. The linear velocity of arm with respect to the UAV in body frame can be determined as:

$$\dot{\boldsymbol{p}}_{\mathrm{Bs}}^{\mathrm{B}} = \boldsymbol{J}_{\mathrm{p}}(s)\, \dot{\boldsymbol{q}}. \tag{9}$$

Here, $\boldsymbol{J}_{\mathrm{p}}(s)$ is the position Jacobian matrix for linear velocity of the arm in the UAV body frame.

The rotation matrix relating the arm at section s and UAV body frame is obtained from Eq. (10) and the angular velocity of each section of arm in $\mathcal{F}_{\mathrm{B}}$ is calculated as Eq. (11).

$$\boldsymbol{R}_{\mathrm{S}}^{B} = \begin{bmatrix} \mathrm{C}^2 q_2 \mathrm{C}\left(\frac{sq_1}{l}\right) + \mathrm{S}^2 q_2 & \mathrm{C}q_2\mathrm{S}q_2\mathrm{C}\left(\frac{sq_1}{l}\right) - \mathrm{C}q_2\mathrm{S}q_2 & \mathrm{C}q_2\mathrm{S}\left(\frac{sq_1}{l}\right) \\ \mathrm{C}q_2\mathrm{S}q_2\mathrm{C}\left(\frac{sq_1}{l}\right) - \mathrm{C}q_2\mathrm{S}q_2 & \mathrm{S}^2 q_2 \mathrm{C}\left(\frac{sq_1}{l}\right) + \mathrm{C}^2 q_2 & \mathrm{S}q_2\mathrm{S}\left(\frac{sq_1}{l}\right) \\ -\mathrm{C}q_2\mathrm{S}\left(\frac{sq_1}{l}\right) & -\mathrm{S}q_2\mathrm{S}\left(\frac{sq_1}{l}\right) & \mathrm{C}\left(\frac{sq_1}{l}\right) \end{bmatrix} = \begin{bmatrix} \boldsymbol{n}_{\mathrm{s}} & \boldsymbol{b}_{\mathrm{s}} & \boldsymbol{t}_{\mathrm{s}} \end{bmatrix}, \tag{10}$$

$$\boldsymbol{\omega}_{\mathrm{s}}^{\mathrm{B}} = \mathrm{Skew}(t_{\mathrm{s}})\, \dot{t}_{\mathrm{s}} = \boldsymbol{J}_o(s)\, \dot{\boldsymbol{q}}, \tag{11}$$

where $\boldsymbol{n}_{\mathrm{s}}$, $\boldsymbol{b}_{\mathrm{s}}$ and $\boldsymbol{t}_{\mathrm{s}}$ are columns of $\boldsymbol{R}_{\mathrm{S}}^{B}$. Here, $\boldsymbol{J}_o(s)$ is Jacobian matrix for angular velocity of the arm in the UAV body frame. The Jacobian matrices are calculated as:

$$\boldsymbol{J}_{\mathrm{p}}(s) = \begin{bmatrix} l\mathrm{C}q_2 \frac{\mathrm{C}\left(\frac{sq_1}{l}\right)-1}{q_1^2} + s\mathrm{S}\left(\frac{sq_1}{l}\right)\frac{\mathrm{C}q_2}{q_1} & l\mathrm{S}q_2 \frac{\mathrm{C}\left(\frac{sq_1}{l}\right)-1}{q_1} \\ l\mathrm{S}q_2 \frac{\mathrm{C}\left(\frac{sq_1}{l}\right)-1}{q_1^2} + s\mathrm{S}\left(\frac{sq_1}{l}\right)\frac{\mathrm{S}q_2}{q_1} & -l\mathrm{C}q_2 \frac{\mathrm{C}\left(\frac{sq_1}{l}\right)-1}{q_1} \\ -l\frac{\mathrm{S}\left(\frac{sq_1}{l}\right)}{q_1^2} + s\frac{\mathrm{C}\left(\frac{sq_1}{l}\right)}{q_1} & 0 \end{bmatrix}, \tag{12}$$

$$\boldsymbol{J}_o(s) = \begin{bmatrix} -\frac{s}{l}\mathrm{S}q_2 & -\mathrm{C}\left(\frac{sq_1}{l}\right)\mathrm{S}\left(\frac{sq_1}{l}\right)\mathrm{C}q_2 \\ \frac{s}{l}\mathrm{C}q_2 & -\mathrm{C}\left(\frac{sq_1}{l}\right)\mathrm{S}\left(\frac{sq_1}{l}\right)\mathrm{S}q_2 \\ 0 & \mathrm{S}^2\left(\frac{sq_1}{l}\right) \end{bmatrix}. \tag{13}$$

By taking derivate of Eq. (7), the linear velocity of arm in $\mathcal{F}_{\mathrm{I}}$ can be computed by Eq. (14).

$$\dot{\boldsymbol{p}}_{\mathrm{s}} = \dot{\boldsymbol{p}}_{\mathrm{B}} - \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{B}\right)\boldsymbol{\omega}_{\mathrm{B}} + \boldsymbol{R}_{\mathrm{B}}\boldsymbol{J}_{\mathrm{p}}(s)\,\dot{\boldsymbol{q}}. \tag{14}$$

Also, the angular velocity of arm for each section in the inertial frame can be expressed as

$$\boldsymbol{\omega}_{\mathrm{s}} = \boldsymbol{\omega}_{\mathrm{B}} + \boldsymbol{R}_{\mathrm{B}}\boldsymbol{J}_{\mathrm{o}}(s)\,\dot{\boldsymbol{q}}. \tag{15}$$

After formulating kinematics of ACMS, the dynamic model needs to be derived. To this end, Euler-Lagrange formulation is adopted. The Lagrangian, L, is obtained by formulating the total kinetic and potential energies, K, and U, respectively, as follows:

$$\mathrm{L} = \mathrm{K} - \mathrm{U}. \tag{16}$$

The Euler-Lagrange equation is then utilized as:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\boldsymbol{X}}_{\mathrm{i}}} - \frac{\partial L}{\partial \boldsymbol{X}_{\mathrm{i}}} + \frac{\partial D}{\partial \dot{\boldsymbol{X}}_{\mathrm{i}}} = u_{\mathrm{i}}, \qquad \mathrm{i} = 1, \ldots, \mathrm{n}_{\xi}, \tag{17}$$

where $\boldsymbol{X}_{\mathrm{i}}$ is the ith generalized coordinate while u_{i} is the ith generalized force and D is the viscous damper which is assumed to be linear.

Total kinetic energy, K, consists of UAV kinetic energy, K_{B} and continuum arm kinetic energy (main backbone, K_{mb}, and disks, K_{d}).

The kinetic energy of UAV is determined as Eq. (18) while the kinetic energy of main backbone for each section is presented in Eq. (19). To determine the total kinetic energy for the backbone, we should integrate Eq. (19) along the length of continuum arm.

$$K_{\mathrm{B}} = \frac{1}{2}m_{\mathrm{B}}\dot{\boldsymbol{p}}_{B}^{\mathrm{T}}\dot{\boldsymbol{p}}_{B} + \frac{1}{2}\boldsymbol{\omega}_{B}^{\mathrm{T}}\boldsymbol{R}_{B}\boldsymbol{I}_{B}\boldsymbol{R}_{B}^{\mathrm{T}}\boldsymbol{\omega}_{B} = \frac{1}{2}m_{B}\dot{\boldsymbol{p}}_{B}^{\mathrm{T}}\dot{\boldsymbol{p}}_{\mathrm{B}} + \frac{1}{2}\dot{\boldsymbol{\phi}}_{B}^{\mathrm{T}}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{I}_{B}\boldsymbol{Q}\,\dot{\boldsymbol{\phi}}_{\mathrm{B}}, \tag{18}$$

where m_B is the mass of UAV and $\boldsymbol{I}_{\mathrm{B}}$ denotes the UAV moment of inertia matrix in body frame.

$$K_{\mathrm{mb}}(s) = \frac{1}{2}\rho_{\mathrm{s}}A_{\mathrm{s}}\dot{\boldsymbol{p}}_{\mathrm{s}}^{\mathrm{T}}\dot{\boldsymbol{p}}_{\mathrm{s}} + \frac{1}{2}\boldsymbol{\omega}_{\mathrm{s}}^{\mathrm{T}}\boldsymbol{R}_{\mathrm{B}}\boldsymbol{R}_{\mathrm{s}}^{\mathrm{B}}\boldsymbol{I}_{\mathrm{s}}\boldsymbol{R}_{\mathrm{s}}^{\mathrm{BT}}\boldsymbol{R}_{\mathrm{B}}^{\mathrm{T}}\boldsymbol{\omega}_{\mathrm{s}}. \tag{19}$$

Here, ρ_{s} indicates the density of main backbone while A_{s} and $\boldsymbol{I}_{\mathrm{s}}$ are the cross-section and moment of inertia for each section of the main backbone. From Eq. (20), the kinetic energy for one disk of continuum arm is obtained.

$$K_{\mathrm{d}}(\mathrm{j}) = \frac{1}{2}m_{\mathrm{d}}\dot{\boldsymbol{p}}_{\mathrm{d}}^{\mathrm{T}}\dot{\boldsymbol{p}}_{\mathrm{d}} + \frac{1}{2}\boldsymbol{\omega}_{\mathrm{d}}^{\mathrm{T}}\boldsymbol{R}_{\mathrm{B}}\boldsymbol{R}_{\mathrm{d}}^{\mathrm{B}}\boldsymbol{I}_{\mathrm{d}}\boldsymbol{R}_{\mathrm{d}}^{\mathrm{BT}}\boldsymbol{R}_{\mathrm{B}}^{T}\boldsymbol{\omega}_{\mathrm{d}}. \tag{20}$$

Here, position of each disk and disk's rotational velocity are, $\boldsymbol{p}_{\mathrm{d}} = \boldsymbol{p}_{\mathrm{s}}\big|_{\mathrm{s=jh}}$, $\mathrm{j} = 1, 2, 3, \ldots,\ n$, and $\boldsymbol{\omega}_{\mathrm{d}} = \boldsymbol{\omega}_{\mathrm{s}}|_{\mathrm{s=jh}}$, respectively, where n is the number of disks and h denotes the equal distance between disks. Moreover, $\boldsymbol{I}_{\mathrm{d}}$ indicates the moment of inertia for each disk.

Finally, total kinetic energy, K, can be calculated from Eqs. (18)–(20) as Eq. (21) which can be rewritten in the compact form as follows.

$$\mathrm{K} = K_{\mathrm{B}} + K_{\mathrm{mb}} + K_{\mathrm{d}} = K_{\mathrm{B}} + \int_{0}^{l} K_{\mathrm{mb}}(s)\,ds + \sum_{\mathrm{j}=1}^{n} K_{\mathrm{d}}(\mathrm{j}), \tag{21}$$

$$\mathrm{K} = \frac{1}{2}\dot{\boldsymbol{X}}^{\boldsymbol{T}}\boldsymbol{M}\dot{\boldsymbol{X}}, \tag{22}$$

where elements of $\boldsymbol{M}$ matrix can be found in the Appendix.

The potential energy for the UAV is calculated as Eq. (23), while the contribution of main backbone and disks are given in Eq. (24) and Eq. (25), respectively.

$$U_{\mathrm{B}} = m_{\mathrm{B}}g\boldsymbol{e}_{3}^{\mathrm{T}}\boldsymbol{p}_{\mathrm{B}}, \tag{23}$$

$$U_{\mathrm{mb}} = \int_{0}^{l}\left(\rho_{\mathrm{s}}A_{\mathrm{s}}g\boldsymbol{e}_{3}^{\mathrm{T}}\left(\boldsymbol{p}_{\mathrm{B}} + \boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)\right)ds + \frac{2E_{\mathrm{s}}\boldsymbol{I}_{\mathrm{s}}}{l}q_{1}^{2}, \tag{24}$$

$$U_{\mathrm{d}} = \sum_{\mathrm{j}=1}^{n}\left(m_{\mathrm{d}}(\mathrm{j})\,g\boldsymbol{e}_{3}^{\mathrm{T}}\left(\boldsymbol{p}_{\mathrm{B}} + \boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)\right)\Big|_{\mathrm{s=jh}}, \tag{25}$$

where g is the gravity acceleration and $\boldsymbol{e}_{3} = [\,0 \quad 0 \quad 1\,]^{T}$ while E_{s} denotes Young's modulus of the main backbone. Finally, the total potential energy of ACMS can be determined as

$$\mathrm{U} = U_{\mathrm{B}} + U_{\mathrm{mb}} + U_{\mathrm{d}}. \tag{26}$$

Consequently, the whole dynamic of ACMS can be written as

$$\boldsymbol{M}(\boldsymbol{X})\,\ddot{\boldsymbol{X}} + \boldsymbol{C}\left(\boldsymbol{X}, \dot{\boldsymbol{X}}\right)\dot{\boldsymbol{X}} + \boldsymbol{G}(\boldsymbol{X}) = \boldsymbol{u} + \boldsymbol{u}_{\boldsymbol{ext}}, \tag{27}$$

where $\boldsymbol{G}(\boldsymbol{X})$ and the elements of matrix $\boldsymbol{C}$ are determined as

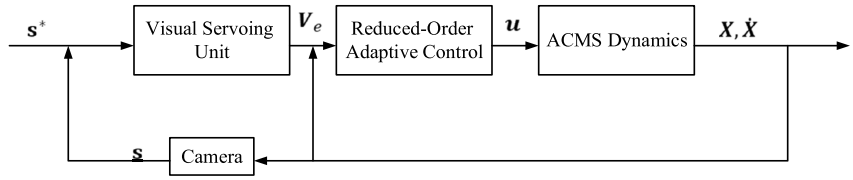

Fig. 2. Vision-based reduced order adaptive control block diagram.

$$\boldsymbol{G}(\boldsymbol{X}) = \left(\frac{\partial \mathrm{U}(\boldsymbol{X})}{\partial \boldsymbol{X}}\right)^{\boldsymbol{T}},$$

$$\boldsymbol{C_{ij}} = \frac{\boldsymbol{1}}{\boldsymbol{2}}\sum_{\boldsymbol{k=1}}^{8}\left(\frac{\partial \boldsymbol{M_{ij}}}{\partial \boldsymbol{X_k}} + \frac{\partial \boldsymbol{M_{ik}}}{\partial \boldsymbol{X_j}} + \frac{\partial \boldsymbol{M_{jk}}}{\partial \boldsymbol{X_i}}\right)\dot{\boldsymbol{X}}_{\boldsymbol{k}}, \qquad \mathrm{i, j} = 1, \ldots, 8.$$

In Eq. (27), $\boldsymbol{u}$ is the generalized force and is defined in Eq. (28) while $\boldsymbol{u_{ext}}$ is the external force.

$$\boldsymbol{u} = \mathrm{diag}(\boldsymbol{R}_{\mathrm{B}}, \boldsymbol{Q}^{\mathrm{T}}, \boldsymbol{I_2})\mathrm{diag}(\boldsymbol{\Omega}, \boldsymbol{I_2})\left[\begin{array}{cc} \boldsymbol{f}_B^T & \boldsymbol{f}_{ten}^T \end{array}\right]^{\mathrm{T}}, \tag{28}$$

where $\boldsymbol{f}_B^T$ is the input vector of forces given by the UAV motors and $\boldsymbol{f}_{ten}^T$ represents the generalized force for continuum arm which can be found based on tendon actuation in [26]. Furthermore, $\boldsymbol{I_2}$ is identity matrix of size 2 and $\boldsymbol{\Omega}$ can be calculated based on motors configurations [27]. In the case of having an under-actuated platform, $\boldsymbol{\Omega}$ can be found as:

$$\boldsymbol{\Omega} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & d & 0 & -d \\ -d & 0 & d & 0 \\ c & -c & c & -c \end{bmatrix}.$$

Here d is the distance from motors to the center of the UAV and c is the drag term.

3. Control architecture

As it was explained earlier, due to the inherent nature of ACMS's structure and its operational circumstance, a high level of uncertainties is expected. As noted, dynamic model of ACMS introduces significant computational challenges and even with significant calibration efforts, model uncertainties may prevail. Furthermore, the existence of various environmental disturbances may further increase the level of positioning uncertainties and degrade the positioning control performance. When automated positioning with respect to an object is concerned, the target object position may also introduce further uncertainties. As a remedy to the positioning problem, a cascaded control approach is proposed to enable automated end-point positioning in unstructured environments. As it is shown in Fig. 2, in the inner loop a reduced-order adaptive control resides to reduce control dependence on ACMS dynamics knowledge. In the outer-loop a vision-based control method is provided to enable direct control of the end-point with respect to an object (target), reducing susceptibility of the positioning task to position uncertainties of the ACMS and object. Design of each of the involved control methods will be explained in the coming sections.

4. Reduced-ordered adaptive control

As illustrated in Fig. 2, outer visual servoing loop is responsible to calculate feature error and generate desired velocity of ACMS parameters for the inner reduced order adaptive control loop. In the inner loop, adaptive control estimates system parameters and calculates control signal to follow commands of the outer loop.

Since the dynamics of ACMS is highly nonlinear, designing appropriate control algorithm is quite challenging. To overcome this problem, discrete-time reduced-order adaptive scheme which is a non-model-based control approach is proposed for ACMS control. In this method, coupled nonlinearities in the system dynamics are treated as separate terms during the control process which enables us to control each axis separately. In the proposed control method, dynamics of ACMS is formulated in form of decoupled equations, then the resulted continuous time equations are discretized to facilitate control design. In the next step, parameters of discrete-time model are estimated for adaptive control purpose.

To determine decoupled equations of ACMS, the dynamic equation of our system from Eq. (27) can be expressed as

$$\begin{bmatrix} M_{11} & \cdots & M_{18} \\ \vdots & \ddots & \vdots \\ M_{81} & \ldots & M_{88} \end{bmatrix}\ddot{\boldsymbol{X}} = \boldsymbol{u}(t) + \boldsymbol{u_{ext}}(t) - \boldsymbol{g}(\boldsymbol{X}, \dot{\boldsymbol{X}}), \tag{29}$$

where $\boldsymbol{g}(\boldsymbol{X}, \dot{\boldsymbol{X}}) = \boldsymbol{C}\left(\boldsymbol{X}, \dot{\boldsymbol{X}}\right)\dot{\boldsymbol{X}} + \boldsymbol{G}(\boldsymbol{X})$. By diagonalizing matrix $\mathbf{M}$ and rewriting above equation, we have

$$\begin{bmatrix} M_{11} & & 0 \\ & \ddots & \\ 0 & & M_{88} \end{bmatrix}\ddot{\boldsymbol{X}} = \boldsymbol{u}(t) + \boldsymbol{u_{ext}}(t) - \boldsymbol{g}\left(\boldsymbol{X}, \dot{\boldsymbol{X}}\right) - \boldsymbol{\delta}(\boldsymbol{X}, \ddot{\boldsymbol{X}}), \tag{30}$$

where $\boldsymbol{\delta}(\boldsymbol{X}, \ddot{\boldsymbol{X}})$ represents the coupling terms between axes and is described as

$$\boldsymbol{\delta}(\boldsymbol{X}, \ddot{\boldsymbol{X}}) = \begin{bmatrix} 0 & & M_{ij} \\ & \ddots & \\ M_{ji} & & 0 \end{bmatrix} \ddot{\boldsymbol{X}}, i = 1, \ldots, 8 \text{ (row number) and } \quad j = 1, \ldots, 8 \text{ (column number)}. \tag{31}$$

Therefore, the dynamic model of ACMS can be written in the state-space form as

$$\boldsymbol{Y} = \begin{bmatrix} \boldsymbol{X} \\ \dot{\boldsymbol{X}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \end{bmatrix}, \tag{32}$$

$$\dot{\boldsymbol{Y}} = \begin{bmatrix} \dot{\boldsymbol{X}} \\ \ddot{\boldsymbol{X}} \end{bmatrix} = \begin{bmatrix} \dot{\boldsymbol{X}} \\ \boldsymbol{f}(\boldsymbol{Y}(t), \boldsymbol{u}(t), t) \end{bmatrix}, \tag{33}$$

$$f_i(\boldsymbol{Y}(t), \boldsymbol{u}(t), t) = (M_{ii})^{-1} \left(u_i(t) - \psi_i\left(\boldsymbol{X}, \dot{\boldsymbol{X}}, \ddot{\boldsymbol{X}}\right)\right), \quad i = 1, ..., 8, \tag{34}$$

where ψ_i is the term represents the external disturbances, unmodeled dynamics and coupling nonlinearities, and is described as

$$\psi_i(\boldsymbol{X}, \dot{\boldsymbol{X}}, \ddot{\boldsymbol{X}}) = -u_{exti}(t) + g_i(\boldsymbol{X}, \dot{\boldsymbol{X}}) + \delta_i(\boldsymbol{X}, \ddot{\boldsymbol{X}}). \tag{35}$$

By defining ψ_i in Eq. (35) and using Eq. (30), the dynamic equation for each axis can be formulated as

$$M_{ii}\ddot{\boldsymbol{X}}_i = u_i(t) - \psi_i(\boldsymbol{X}, \dot{\boldsymbol{X}}, \ddot{\boldsymbol{X}}). \tag{36}$$

Next step in control design is discretization of continuous time model. In this regard, the continuous time state-space form of ACMS derived in Eq. (33) can be discretized using discrete Taylor expansion as Eqs. (37)–(38) with the assumption that changes in M_{ii} and $\psi_i(\boldsymbol{X}, \dot{\boldsymbol{X}}, \ddot{\boldsymbol{X}})$ are negligible during the sampling time T. It should be noted that selecting large sampling time values can affect performance and efficiency of the controller.

$$y_{1i}(k) = y_{1i}(k-1) + T y_{2i}(k-1) + 0.5\left(\frac{T^2}{M_{ii}}\right)(u_i(k-1) - \psi_i(k-1)), \tag{37}$$

$$y_{2i}(k) = y_{2i}(k-1) + \left(\frac{T}{M_{ii}}\right)(u_i(k-1) - \psi_i(k-1)). \tag{38}$$

Here, k is the sampling instant. Finally, the discretized dynamics of ACMS can be described as:

$$\left(1 - q^{-1}\right) y_{2i}(k) = q^{-2}\left(b_i u_i(k) - \acute{\psi}_i(k+1)\right), \tag{39}$$

$$b_i = \left(\frac{T}{M_{ii}}\right), \tag{40}$$

$$\acute{\psi}_i(k) = \left(\frac{T}{M_{ii}}\right)\psi(k-1), \tag{41}$$

where q^{-1} is the backward shift operator. Considering that the elements of Eq. (39) are unknown, to design a controller without having any knowledge of the system dynamics, the Eq. (39) can be rewritten as:

$$\left(1 - q^{-1}\right) y_{2i}(k) = \hat{\theta}_i \phi_i(k-1), \tag{42}$$

$$\hat{\theta}_i = \begin{bmatrix} \hat{b}_i & \widehat{\acute{\psi}}_i \end{bmatrix}, \qquad \phi_i^T(k) = \begin{bmatrix} u_i(k) & 1 \end{bmatrix}. \tag{43}$$

Here, $\hat{\theta}_i$ is the estimation of parameters in the system dynamics and can be obtained by the recursive parameter adaptation algorithm.

$$\hat{\theta}_i(k) = \hat{\theta}_i(k-1) + F_i(k)\phi_i(k-2)\varepsilon_i^0(k), \tag{44}$$

where $F_i(k)$ and $\varepsilon_i^0(k)$ are calculated as follows [15].

$$\varepsilon_i^0(k) = \frac{y_{2i}(k) - y_{2i}(k-1) - \hat{\theta}_i^{\mathrm{T}}(k-1)\phi_i(k-2)}{1 + \phi_i^{\mathrm{T}}(k-2)F_i(k-1)\phi_i(k-2)},$$

$$F_i(k+1) = \frac{1}{\lambda_1(k)}\left[F_i(k) - \frac{F_i(k)\phi_i(k-2)\phi_i^{\mathrm{T}}(k-2)F_i(k)}{\frac{\lambda_1(k)}{\lambda_2(k)} + \phi_i^{\mathrm{T}}(k-2)F_i(k-1)\phi_i(k-2)}\right],$$

$$0 < \lambda_1 \le 1, 0 < \lambda_2 \le 2, F_i(0) > 0.$$

Having estimated parameter of the system, the control signal can be designed as [15]:

$$u_i(k) = -\iota_i u_i(k-1) + \frac{1}{\hat{b}_i(k)}\left[\gamma_i\left(q^{-1}\right) y_{2d}(k+2) - \rho_i y_{2i}(k) + \widehat{\acute{\psi}}_i\right],$$

$$\gamma_i\left(q^{-1}\right) = \left(1 - q^{-1}\right)\Lambda_i\left(q^{-1}\right) + q^{-2}\rho_i,$$

$$\Lambda_i\left(q^{-1}\right) = 1 + \iota_i q^{-1}, \tag{45}$$

where y_{2d} is desired values of y_2 whereas ι_i and ρ_i are positive constant parameters.

The stability of the proposed controller can be found in [28,29] based on hyperstability theory.

5. Visual servoing

In this section, a vision-based control method of aerial manipulation is discussed for both under-actuated and fully-actuated base platforms. Using visual feed back shown in Fig. 2, desired states for reduced adaptive control method are obtained. Here, we use eye-in-hand configuration for visual servoing, where camera is attached to the end-effector of a continuum arm. For the reasons expressed earlier, an IBVS approach is adopted.

Since visual servoing commands are calculated in camera frame, we need to derive the end-effector velocity in camera frame. In this regard, using Eqs. (14) and (15) expressed in inertia frame, end-effector velocity, $\boldsymbol{V}_e$, in camera frame can be obtained as

$$\boldsymbol{V}_e = \left(\boldsymbol{R}_B \boldsymbol{R}_{\mathrm{s}}^{\mathrm{B}}\right)^{-1} \begin{bmatrix} \dot{\boldsymbol{p}}_{\mathrm{s}} \\ \boldsymbol{\omega}_{\mathrm{s}} \end{bmatrix}_{\mathrm{s}=l}. \tag{46}$$

Alternatively, end-effector velocity can be written in the following compact form:

$$\boldsymbol{V}_e = \begin{bmatrix} \boldsymbol{J}_1 & \boldsymbol{J}_2 \end{bmatrix} \begin{bmatrix} \boldsymbol{V}_B \\ \dot{\boldsymbol{q}} \end{bmatrix} = \boldsymbol{J}\boldsymbol{V}. \tag{47}$$

Here, $\boldsymbol{V}_B = \begin{bmatrix} \dot{\boldsymbol{p}}_{\mathrm{B}}^{\mathrm{B}} \\ \boldsymbol{\omega}_{\mathrm{B}}^{\mathrm{B}} \end{bmatrix}$ is the UAV velocity in UAV body frame and $\boldsymbol{J} = \begin{bmatrix} \boldsymbol{J}_1 & \boldsymbol{J}_2 \end{bmatrix}$ is the Jacobian matrix where

$$\boldsymbol{J}_1 = \begin{bmatrix} \boldsymbol{R}_{\mathrm{B}}^{\mathrm{s}} & -\boldsymbol{R}_{\mathrm{B}}^{\mathrm{s}} \boldsymbol{R}_{\mathrm{B}}^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}} \boldsymbol{p}_{\mathrm{Bs}}^{B}\right) \boldsymbol{R}_{\mathrm{B}} \\ 0_{3\times 3} & \boldsymbol{R}_{\mathrm{B}}^{\mathrm{s}} \end{bmatrix}_{\mathrm{s}=l}, \tag{48}$$

$$\boldsymbol{R}_{\mathrm{B}}^{\mathrm{s}} = \left(\boldsymbol{R}_{\mathrm{s}}^{\mathrm{B}}\right)^{-1}, \tag{49}$$

$$\boldsymbol{J}_2 = \begin{bmatrix} \boldsymbol{R}_{\mathrm{B}}^{\mathrm{s}} \boldsymbol{J}_{\mathrm{p}} \\ \boldsymbol{R}_{\mathrm{B}}^{\mathrm{s}} \boldsymbol{J}_{\mathrm{o}} \end{bmatrix}_{s=l}. \tag{50}$$

In case of under-actuated base platform, some states of UAV, e.g., roll and pitch angles cannot be controlled directly. Therefore, Eq. (47) can be divided in two terms including controllable term, $\boldsymbol{J}_c \boldsymbol{V}_c$, and uncontrollable term, $\boldsymbol{J}_{uc} \boldsymbol{V}_{uc}$, as

$$\boldsymbol{V}_e = \boldsymbol{J}\boldsymbol{V} = \boldsymbol{J}_c \boldsymbol{V}_c + \boldsymbol{J}_{uc} \boldsymbol{V}_{uc}. \tag{51}$$

For under-actuated UAVs, hovering mode is a reasonable assumption for aerial manipulation missions applied in many similar aerial manipulation studies [30] because UAV should reach to the feasible area for manipulation purpose. Consequently, roll and pitch angles are negligible. Therefore, uncontrolled term in Eq. (51) does not affect the end effector velocity and Eq. (51) can be approximated as

$$\boldsymbol{V}_e \approx \boldsymbol{J}_c \boldsymbol{V}_c. \tag{52}$$

In under-actuated platform, desired attitude and angular velocity for the inner control loop should be calculated using position control signal. Therefore, control signal in Eq. (27) can be rewritten as $\boldsymbol{u} = \begin{bmatrix} \boldsymbol{u}_1 & \dots & \boldsymbol{u}_8 \end{bmatrix} = \begin{bmatrix} \boldsymbol{u}_{\mathrm{P}}^{\mathrm{T}} & \boldsymbol{u}_{\mathrm{E}}^{\mathrm{T}} & \boldsymbol{u}_{\mathrm{q}}^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}$, where $\boldsymbol{u}_{\mathrm{P}}$, $\boldsymbol{u}_{\mathrm{E}}$ and $\boldsymbol{u}_{\mathrm{q}}$ are control signals for controlling position, attitude and arms of ACMS respectively. Accordingly, one can obtain desired pitch, θ_{d}, and roll angles, φ_{d}, as [14]

$$\begin{aligned}
& u_{\mathrm{P_X}} = \boldsymbol{u}_1 \quad u_{\mathrm{P_Y}} = \boldsymbol{u}_2 \quad u_{\mathrm{P_Z}} = \boldsymbol{u}_3 \\
& \mathbf{f}_{\mathrm{Z}} = \sqrt{u_{\mathrm{P_X}}^2 + u_{\mathrm{P_Y}}^2 + u_{\mathrm{P_Z}}^2}, \\
& \theta_{\mathrm{d}} = \tan^{-1}\left(\frac{\boldsymbol{u}_{\mathrm{P_X}} \cos\psi_{\mathrm{d}} + \boldsymbol{u}_{\mathrm{P_Y}} \sin\psi_{\mathrm{d}}}{\boldsymbol{u}_{\mathrm{P_Z}}}\right), \\
& \varphi_{\mathrm{d}} = \sin^{-1}\left(\frac{\boldsymbol{u}_{\mathrm{P_X}} \sin\psi_{\mathrm{d}} - \boldsymbol{u}_{\mathrm{P_Y}} \cos\psi_{\mathrm{d}}}{\mathbf{f}_{\mathrm{Z}}}\right).
\end{aligned}$$

In case of fully-actuated UAV, Eq. (51) can be used for not only hovering but also all other possible missions such as moving around without any need to simplification.

Vision-based unit is responsible to find $\boldsymbol{V}_e$. Having end-effector velocity from visual servoing, control unit input can be calculated using Eq. (51) or Eq. (52) depends on UAV actuation. It is notable that $\boldsymbol{V}_c$ is defined in body reference frame and we need to transfer it to the inertia frame before feeding to the controller as a desired input.

The only remaining part to complete the control loop is to find appropriate visual feedback, $\boldsymbol{V}_e$, from vision unit. This process will be explained in the rest of this section.

For visual servoing, we use four visual feature points in the camera frame defined as $\boldsymbol{\chi} = [\chi_1, \ \chi_2, \chi_3, \ \chi_{x4} \ ]^T$, where $\chi_i = (X_i, Y_i, Z_i)$ denotes the ith feature point. The main purpose of the vision-based control is to nullify visual error calculated in Eq. (53).

$$\mathbf{e}(t) = \mathbf{s} - \mathbf{s}^*, \tag{53}$$

where $\mathbf{s}^*$ is the desired image vector of target features while $\mathbf{s} = (x, y)$ denotes the currently observed image features vector.

To design a velocity controller, the relationship between the time variation of $\mathbf{s}$ and the camera velocity is obtained as [16]:

$$\dot{\mathbf{e}} = \boldsymbol{L}_e \boldsymbol{V}_e, \tag{54}$$

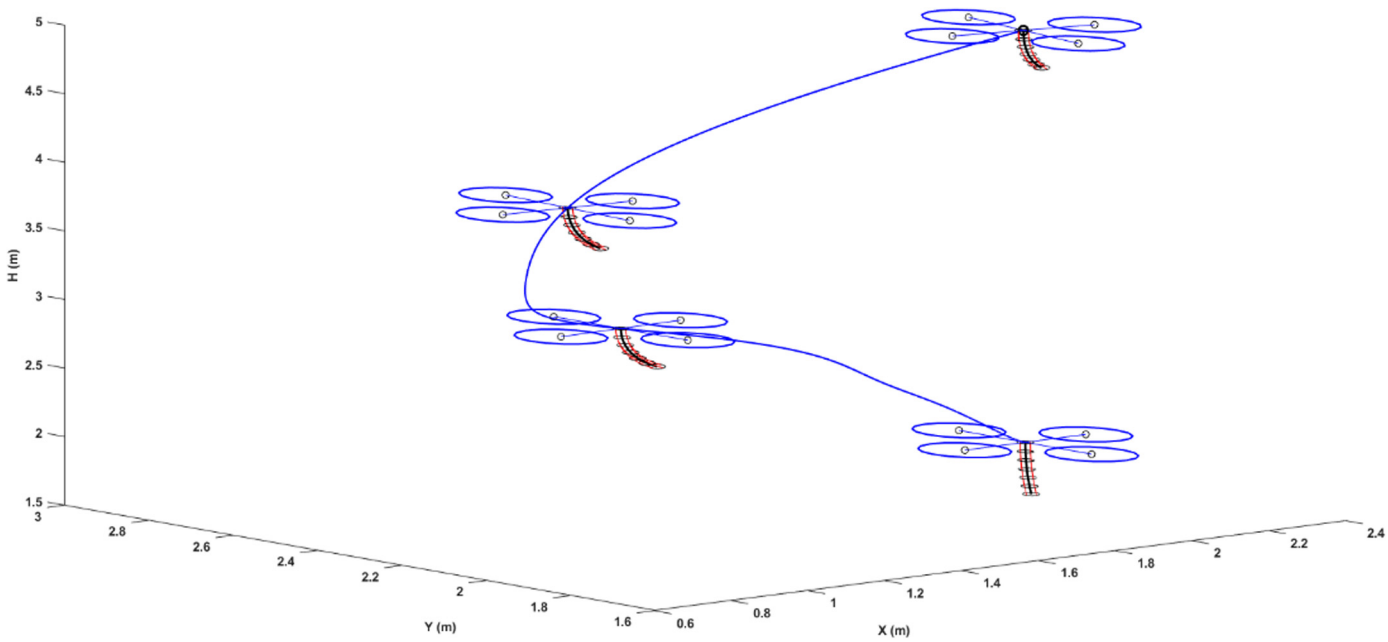

Fig. 3. 3D trajectory of the under-actuated ACMS (first scenario).

where $\boldsymbol{L}_e$ is the so-called interaction matrix which can be found by

$$\boldsymbol{L}_e = \begin{bmatrix} \boldsymbol{L}_{e1} & \boldsymbol{L}_{e2} & \boldsymbol{L}_{e3} & \boldsymbol{L}_{e4} \end{bmatrix}^{\mathrm{T}}, \tag{55}$$

where for point features we will have

$$\boldsymbol{L}_{ei} = \begin{bmatrix} \frac{-1}{Z_i} & 0 & \frac{x_i}{Z_i} & x_i y_i & -\left(1+x_i^2\right) & y_i \\ 0 & \frac{-1}{Z_i} & \frac{y_i}{Z_i} & 1+y_i^2 & -x_i y_i & -x_i \end{bmatrix}, \quad i = 1, \ldots 4. \tag{56}$$

Here, Z_i denotes the depth of feature points relative to the camera, and its value is estimated in the online phase at each iteration, i.e., using Kalman filter during the control process. To guarantee the exponential decrease of error as $\dot{\mathbf{e}} = -\boldsymbol{\lambda}\mathbf{e}$, the final visual servoing control law can be expressed as

$$\boldsymbol{V}_e = -\boldsymbol{\lambda}\boldsymbol{L}_e^{+}\mathbf{e}, \tag{57}$$

where $\boldsymbol{L}_e^{+}$ is the pseudo-inverse of approximation of $\boldsymbol{L}_e$.

6. Simulation results

In this section, the proposed approach for vision-based reduced adaptive control of under-actuated and fully-actuated ACMSs has been tested in a simulator developed in Robotics, Mechatronics, and Automation Laboratory (RMAL). The simulator is based on MATLAB/Simulink environment. An under-actuated quadrotor and a fully-actuated octa-copter UAV with a weight of 2 Kg [31] and 3.2 Kg [32], respectively, endowed with a 40 cm spring steel continuum arm [7] has been used for simulations. The properties of UAVs and arm have been reported in [31,32] and [7], respectively.

A perspective camera is attached to the end-effector with eye-in-hand configuration. The UAV is assumed to start in a hover position with the desired target in the field of view of camera. Visual features include four points corresponding to four vertices of a rectangle. The location of these vertices with respect to the inertial frame are (0.25 −0.25 0) m, (0.25 0.25 0) m, (−0.25 0.25 0) m, and (−0.25 −0.25 0) m. These points are projected onto the image plane with focal length of 1024 pixels and the principal point located at 512 pixels identical for both vertical and horizontal directions.

Since suggested vision-based control approach is applicable for both under-actuated and fully-actuated UAVs, we need to compare the effect of under-actuation in ACMS performance.

For this purpose, behavior of those two types of ACMSs with different platforms is investigated in the first simulation scenario. The initial position of the UAV is located at (2 2 −5) m having Euler angles of (0 0 10) deg while arm angles are $q_1 = 45$ deg and $q_2 = 20$ deg. The desired position of the end-effector is selected as (2 2 −1.6) m. To indicate the robustness of the system, 20% uncertainties and a disturbance in ACMS model are considered.

The 3D trajectory of the under-actuated ACMS for completing the first aerial mission is indicated in Fig. 3.

Fig. 4 presents feature points trajectories with initial points indicated with circle symbols (∘) while their desired points are shown with cross symbol (×). This figure demonstrates that vision-based control for the under-actuated ACMS works properly since feature points reach their desired values. Also, feature parameter errors are shown in Fig. 5. In this figure, it can be seen that after 35 sec the errors have fully converged to zero.

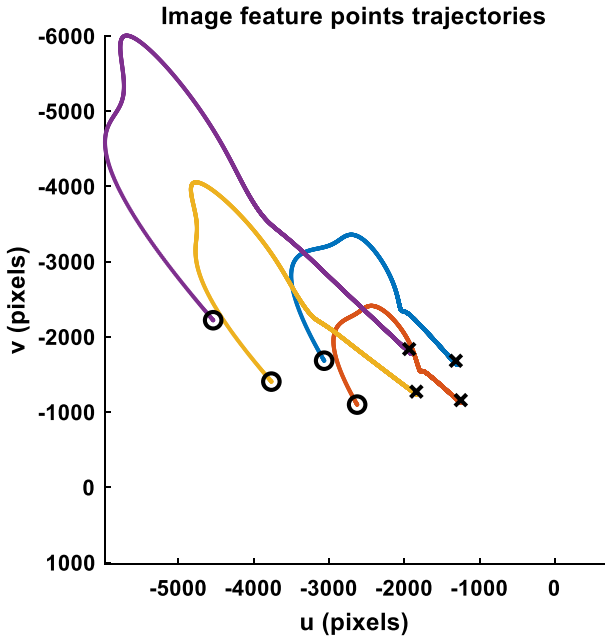

Fig. 4. Feature points trajectories of the under-actuated ACMS (first scenario).

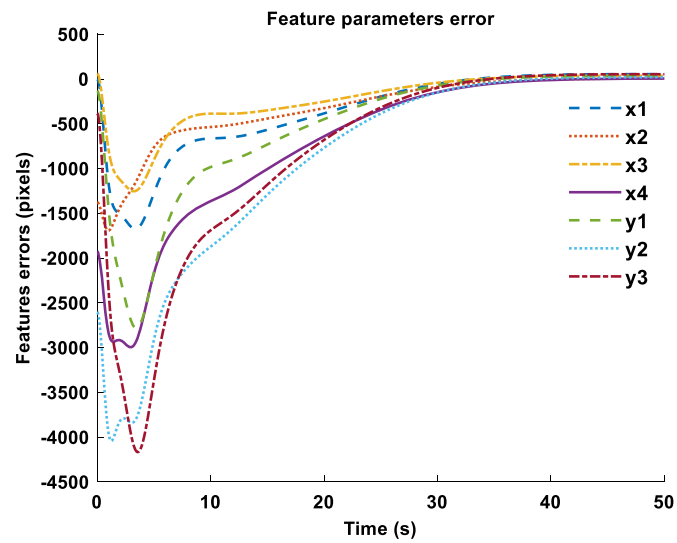

Fig. 5. Feature parameter errors of the under-actuated ACMS (first scenario). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

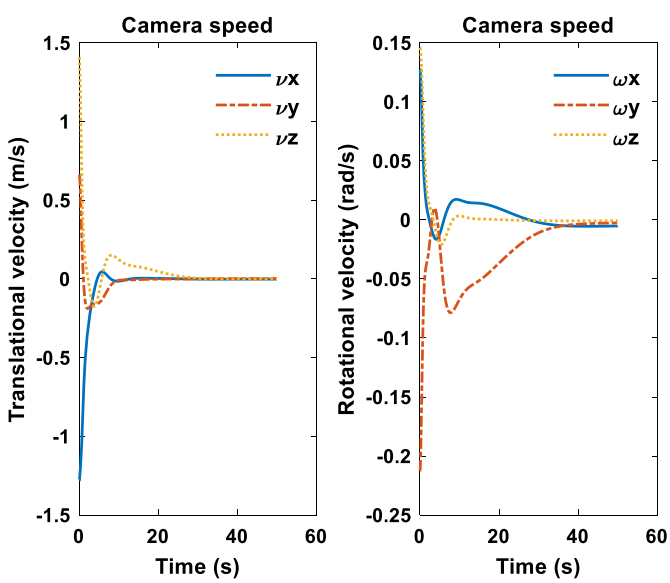

Fig. 6. Camera velocity of the under-actuated ACMS (first scenario).

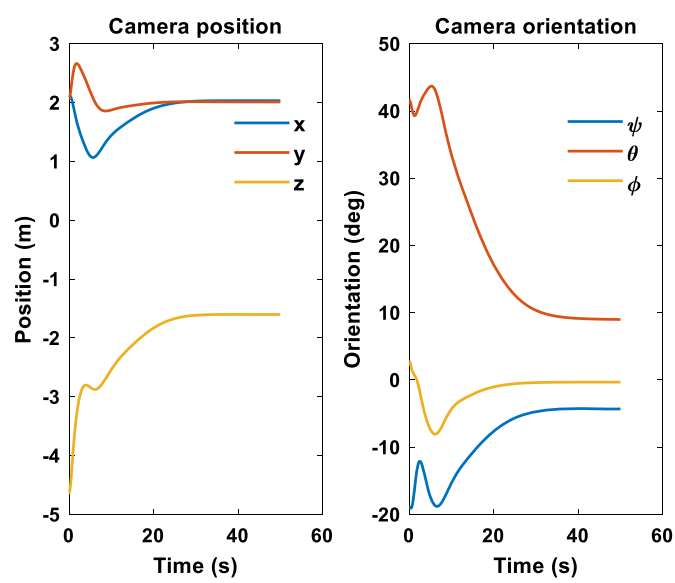

Fig. 7. Camera pose of the under-actuated ACMS (first scenario).

The velocity of end-effector (camera) obtained from vision unit is illustrated in Fig. 6, while the position and orientation of camera with respect to the inertial frame are presented in Fig. 7. Fig. 8 shows the states of ACMS for the trajectory shown in Fig. 3. Finally, Fig. 9 presents control tracking error (error between visual servoing commands and ACMS states) for this scenario.

Similar mission is considered for the fully-actuated ACMS. Results are demonstrated in Figs. 10–16. As shown in Fig. 11, the fully-actuated ACMS experiences a smoother and semi-linear feature point trajectory and a faster feature error convergence compared to the under-actuated ACMS. In the under-actuated ACMS, controller nearly nullifies the Euler angles. Consequently, roll and pitch angles cannot be controlled independently, and desired attitude should be obtained from position control signal while the fully-actuated system enables the independent attitude control for UAV and hence Euler angles converge to different values.

To show the capability of the proposed controller and further investigate the performance of under-actuated and fully-actuated platforms, a complicated mission is chosen for the second scenario. In this scenario, initial positions of the UAV (1 1 −5) m and arm (45, 20) deg are far away from the desired position of the end-effector. Also, initial heading angle is selected 50 deg instead of 10 deg. Other parameters are the same as the first scenario. Also, uncertainties and a disturbance similar to scenario 1 are considered in this scenario.

Because of the limitation of under-actuated ACMS, it fails to complete the mission successfully. Therefore, second scenario results are presented only for the fully-actuated ACMS in Figs. 17–23. Similar to the first scenario, the fully-actuated ACMS demonstrates a fast feature error convergence and its final converged Euler angles are not zero.

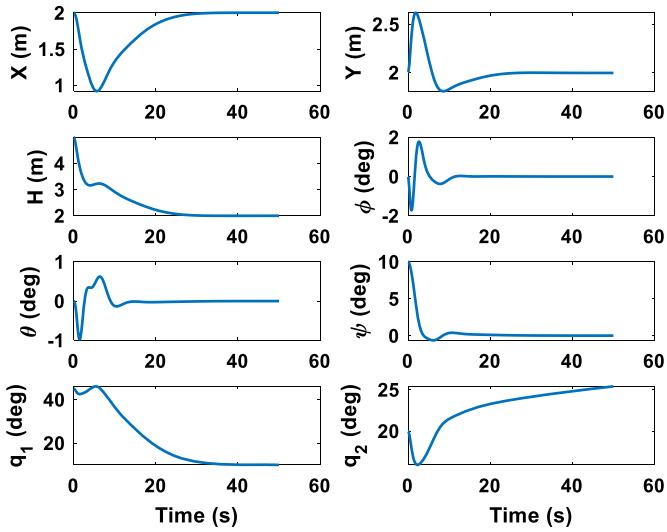

Fig. 8. ACMS states of the under-actuated ACMS (first scenario).

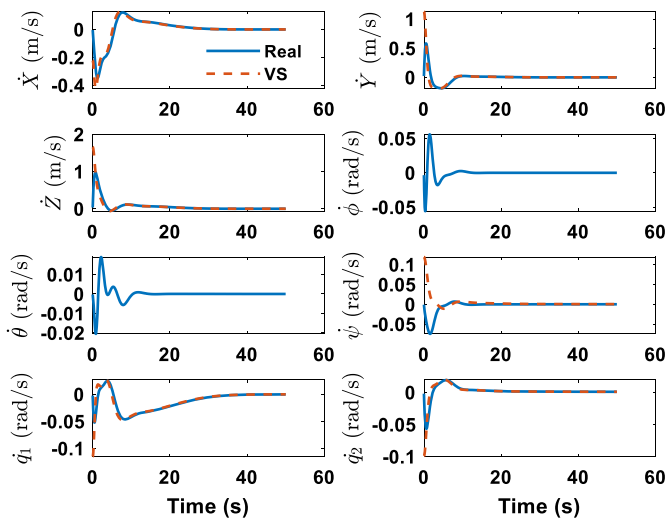

Fig. 9. ACMS tracking error for the under-actuated ACMS (first scenario).

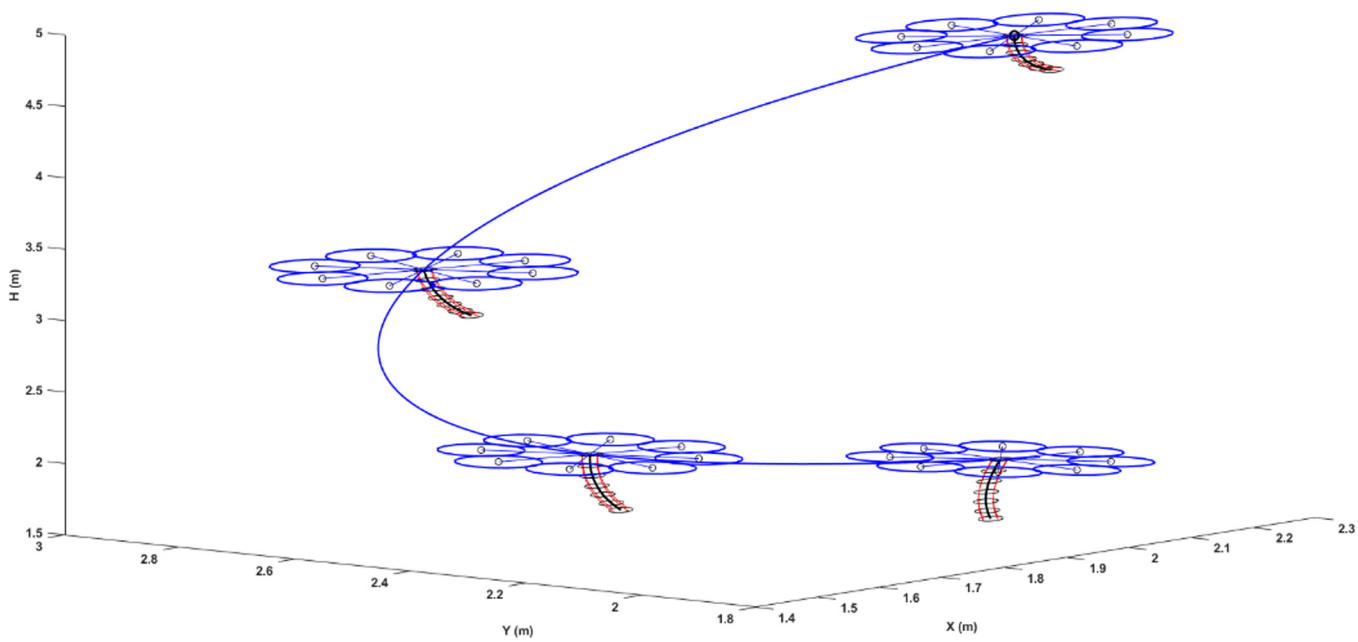

Fig. 10. 3D trajectory of the fully-actuated ACMS (first scenario).

The simulation took 0.04 second per iteration using PC with intel core i7-8700 processor, 3.2 GHz, and 12 GB RAM, within MATLAB 2019b, which is several times faster than model-based execution time in the previous work [7]. This execution time can be reduced up to 500 times in a real-time implementation language such as C [33].

7. Conclusion

This paper contributes by outlining the design of the first vision based adaptive control for ACMSs which circumvents the need for a priori knowledge of system dynamic model and allows robot and target positioning uncertainties. The proposed hybrid controller consists of a reduced-order adaptive controller which does not require the exact dynamic model of the system and estimates the dynamic of the ACMS during the control process. The integrated visual servoing system is also a model-free type, for which an eye-in-hand configuration is adopted. The capability of the proposed controller in learning system model on-the-fly and its robustness to uncertainties and disturbances are verified through the simulation results. The role of UAV actuation in control system is also demonstrated. Our future work will include extending the presented approach in several directions including tackling control problems of compliant motion and manipulation tasks.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

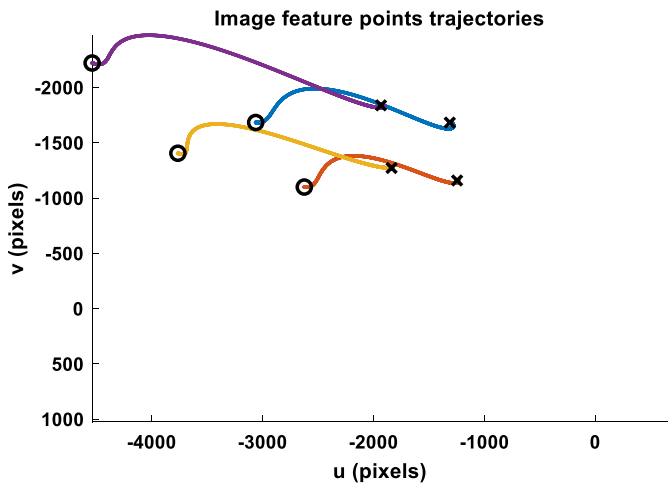

Fig. 11. Feature points trajectories of the fully-actuated ACMS (first scenario).

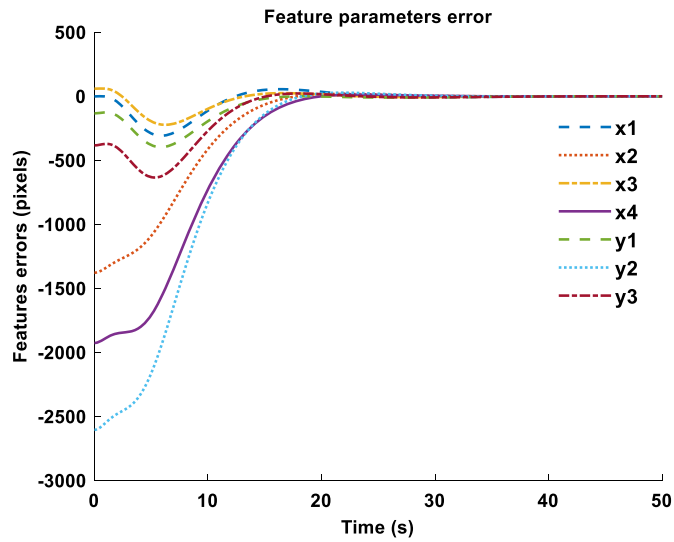

Fig. 12. Feature parameter errors of the fully-actuated ACMS (first scenario).

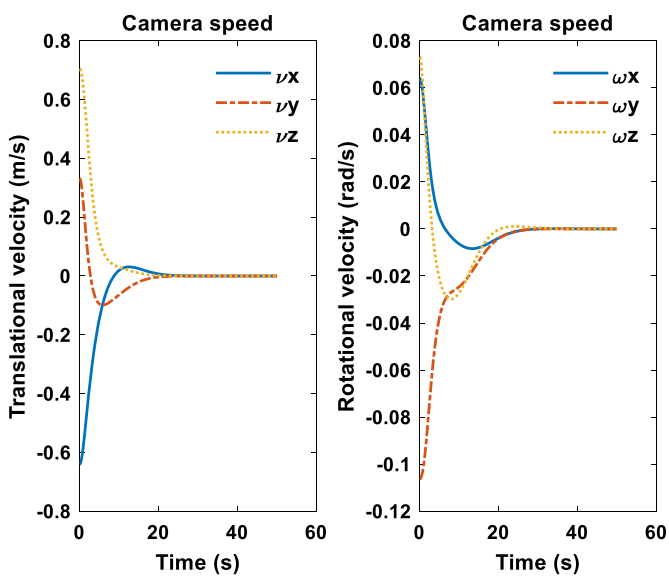

Fig. 13. Camera velocity of the fully-actuated ACMS (first scenario).

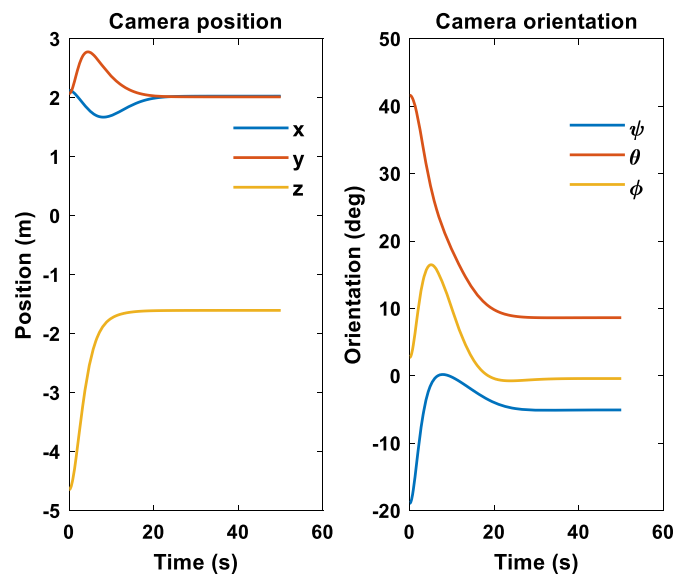

Fig. 14. Camera pose of the fully-actuated ACMS (first scenario).

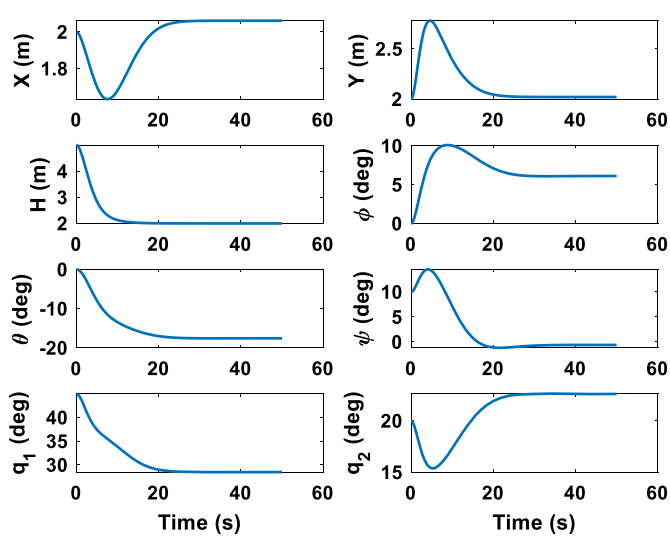

Fig. 15. ACMS states of the fully-actuated ACMS (first scenario).

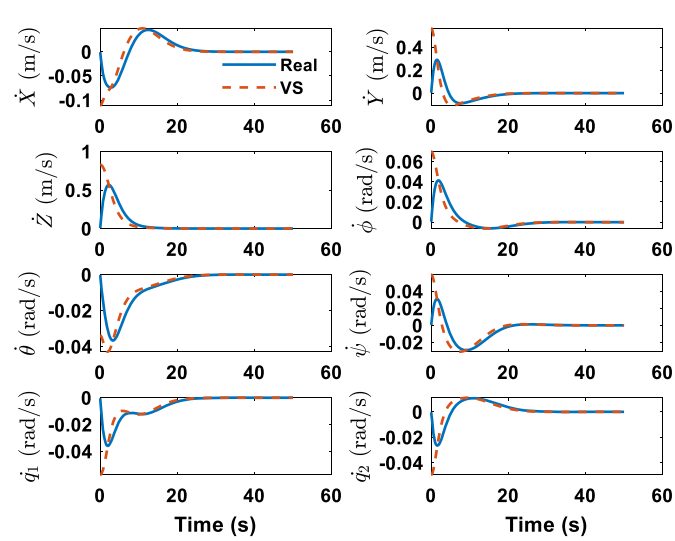

Fig. 16. ACMS tracking error for the fully-actuated ACMS (first scenario).

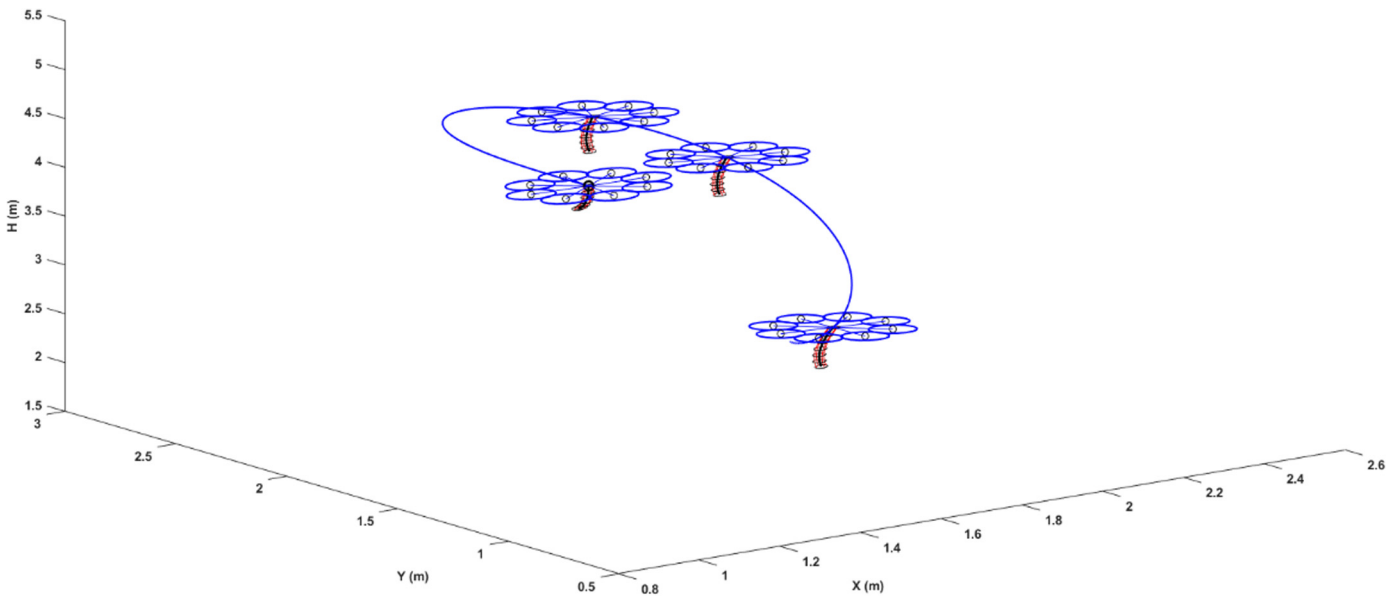

Fig. 17. 3D trajectory of the fully-actuated ACMS (second scenario).

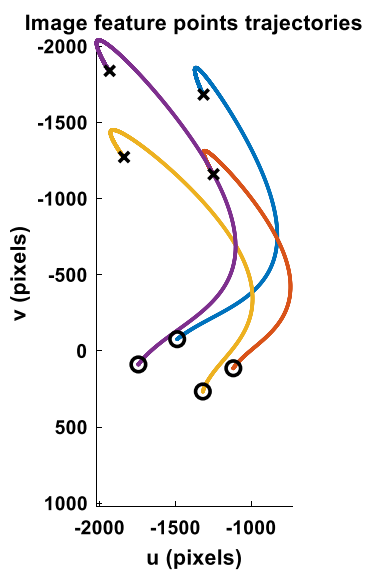

Fig. 18. Feature points trajectories of the fully-actuated ACMS (second scenario).

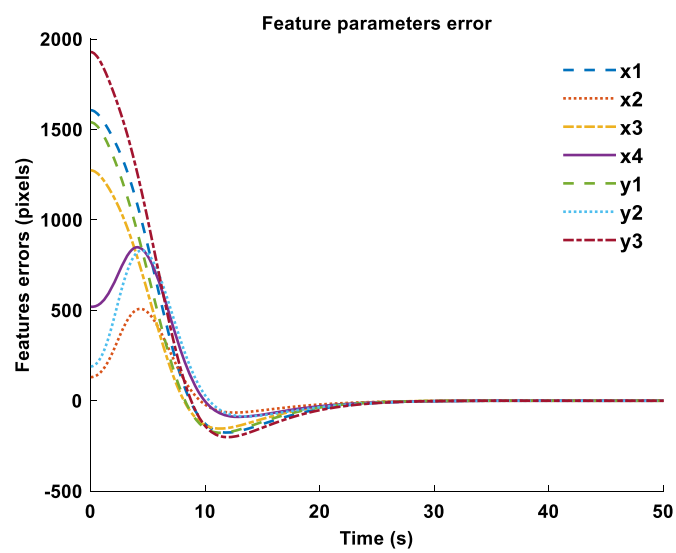

Fig. 19. Feature parameter errors of the fully-actuated ACMS (second scenario).

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Appendix A

The inertial matrix and its elements are presented here.

$$\boldsymbol{M} = \begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} & \boldsymbol{M}_{13} \\ \boldsymbol{M}_{12}^{\mathrm{T}} & \boldsymbol{M}_{22} & \boldsymbol{M}_{23} \\ \boldsymbol{M}_{13}^{\mathrm{T}} & \boldsymbol{M}_{23}^{\mathrm{T}} & \boldsymbol{M}_{33} \end{bmatrix},$$

$$\boldsymbol{M}_{11} = \left(m_{\mathrm{B}} + m_{\mathrm{m.b}} + \sum_{\mathrm{j}=1}^{n} m_{\mathrm{d}}\,(\mathrm{j}) \right) \boldsymbol{I}_{3\times 3},$$

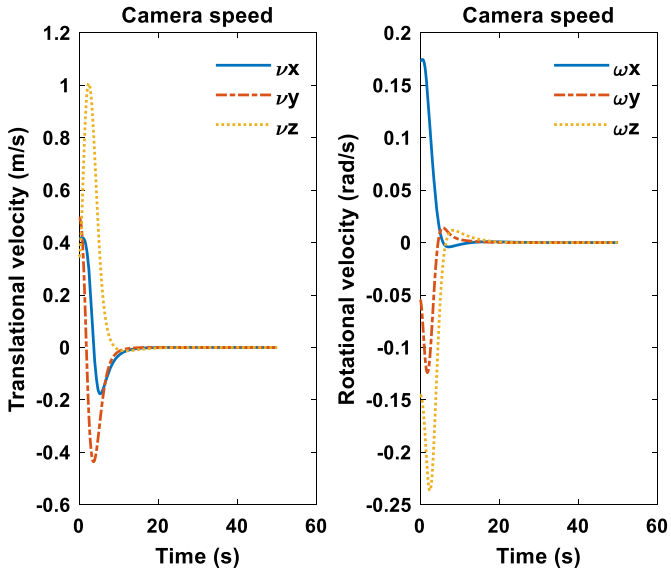

Fig. 20. Camera velocity of the fully-actuated ACMS (second scenario).

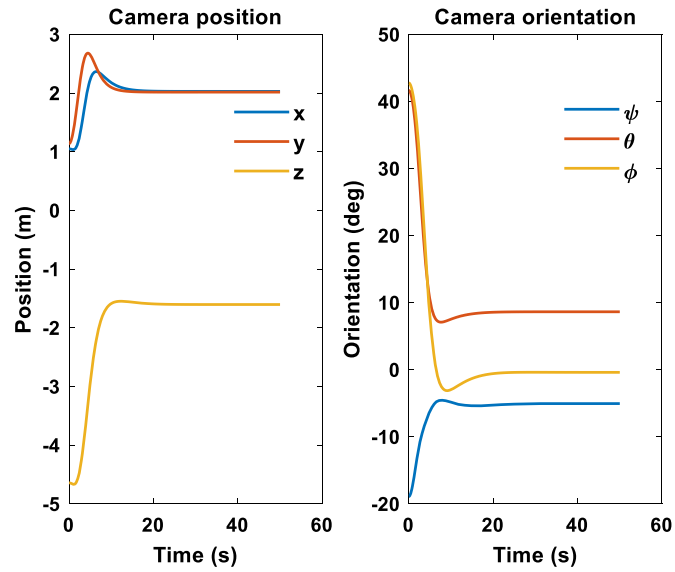

Fig. 21. Camera pose of the fully-actuated ACMS (second scenario).

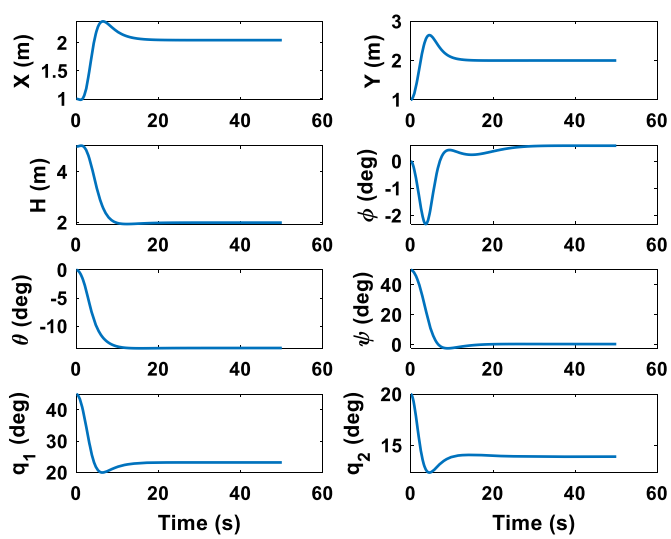

Fig. 22. ACMS states of the fully-actuated ACMS (second scenario).

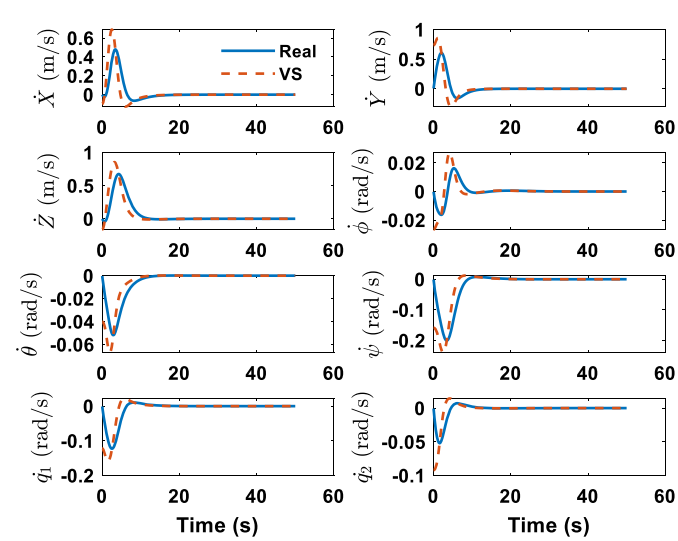

Fig. 23. ACMS tracking error for the fully-actuated ACMS (second scenario).

$$\boldsymbol{M}_{22} = \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{I}_{\mathrm{B}}\boldsymbol{Q} + \int_0^l \left(\rho_{\mathrm{s}} A_{\mathrm{s}} \boldsymbol{T}_{\mathrm{B}}^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)\boldsymbol{T}_{\mathrm{B}} + \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{R}_{\mathrm{S}}^{B}\boldsymbol{I}_{\mathrm{s}}\boldsymbol{R}_{\mathrm{S}}^{B\mathrm{T}}\boldsymbol{Q}\right) ds$$
$$+ \sum_{\mathrm{j}=1}^{n} \left(m_{\mathrm{d}}(\mathrm{j})\, \boldsymbol{T}_{\mathrm{B}}^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)\boldsymbol{T}_{\mathrm{B}} + \boldsymbol{Q}^{\mathrm{T}}\boldsymbol{R}_{\mathrm{S}}^{B}\boldsymbol{I}_{\mathrm{d}}\boldsymbol{R}_{\mathrm{S}}^{B\mathrm{T}}\boldsymbol{Q}\right)\Big|_{\mathrm{s}=\mathrm{jh}},$$

$$\boldsymbol{M}_{33} = \int_0^l \left(\rho_{\mathrm{s}} A_{\mathrm{s}} \boldsymbol{J}_{\mathrm{p}}^{\mathrm{T}}(s)\,\boldsymbol{J}_{\mathrm{p}}(s) + \boldsymbol{J}_{\mathrm{o}}^{\mathrm{T}}(s)\,\boldsymbol{R}_{\mathrm{S}}^{B}\boldsymbol{I}_{\mathrm{s}}\boldsymbol{R}_{\mathrm{S}}^{B\mathrm{T}}\boldsymbol{J}_{\mathrm{o}}(s)\right) ds + \sum_{\mathrm{j}=1}^{n} \left(m_{\mathrm{d}}(\mathrm{j})\, \boldsymbol{J}_{\mathrm{p}}^{\mathrm{T}}(s)\,\boldsymbol{J}_{\mathrm{p}}(s) + \boldsymbol{J}_{\mathrm{o}}^{\mathrm{T}}(s)\,\boldsymbol{R}_{\mathrm{S}}^{B}\boldsymbol{I}_{\mathrm{d}}\boldsymbol{R}_{\mathrm{S}}^{B\mathrm{T}}\boldsymbol{J}_{\mathrm{o}}(s)\right)\Big|_{\mathrm{s}=\mathrm{jh}},$$

$$\boldsymbol{M}_{12} = -\int_0^l \left(\rho_{\mathrm{s}} A_{\mathrm{s}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)\boldsymbol{T}_{\mathrm{B}}\right) ds - \sum_{\mathrm{j}=1}^{n} \left(m_{\mathrm{d}}(\mathrm{j})\, \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}}\boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)\boldsymbol{T}_{\mathrm{B}}\right)\Big|_{\mathrm{s}=\mathrm{jh}},$$

$$\boldsymbol{M}_{13} = \int_0^l \left(\rho_{\mathrm{s}} A_{\mathrm{s}} \boldsymbol{R}_{\mathrm{B}}\boldsymbol{J}_{\mathrm{p}}(s)\right) ds + \sum_{\mathrm{j}=1}^{n} \left(m_{\mathrm{d}}(\mathrm{j})\, \boldsymbol{R}_{\mathrm{B}}\boldsymbol{J}_{\mathrm{p}}(s)\right)\Big|_{\mathrm{s}=\mathrm{jh}},$$

$$\boldsymbol{M}_{23} = \int_0^l \left(\boldsymbol{Q}^{\mathrm{T}} \boldsymbol{R}_{\mathrm{S}}^{B} \boldsymbol{I}_{\mathrm{s}} \boldsymbol{R}_{\mathrm{S}}^{B\mathrm{T}} \boldsymbol{J}_{\mathrm{o}}(s) - \rho_{\mathrm{s}} A_{\mathrm{s}} \boldsymbol{T}_{\mathrm{B}}^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}} \boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)^{\mathrm{T}} \boldsymbol{R}_{\mathrm{B}} \boldsymbol{J}_{\mathrm{p}}(s) \right) ds$$
$$+ \sum_{\mathrm{j}=1}^{n} \left(\boldsymbol{Q}^{\mathrm{T}} \boldsymbol{R}_{\mathrm{S}}^{B} \boldsymbol{I}_{\mathrm{d}} \boldsymbol{R}_{\mathrm{S}}^{B\mathrm{T}} \boldsymbol{J}_{\mathrm{o}}(s) - m_{\mathrm{d}}(j) \boldsymbol{T}_{\mathrm{B}}^{\mathrm{T}} \mathrm{Skew}\left(\boldsymbol{R}_{\mathrm{B}} \boldsymbol{p}_{\mathrm{Bs}}^{\mathrm{B}}\right)^{\mathrm{T}} \boldsymbol{R}_{\mathrm{B}} \boldsymbol{J}_{\mathrm{p}}(s) \right) \Big|_{\mathrm{s}=\mathrm{jh}} .$$

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